



5.3 PRACTICAL EXAMPLES

MIKROELEKTRONIKA

The diagram of a power supply in figure (3.8) uses several diodes. The first four are in a single package, identified by B40C1500. This is a bridge rectifier.

The LED in the circuit indicates the transformer is working. Resistor R1 is used to limit the current through the LED and the brightness of the LED indicates the approximate voltage.

Diodes marked 1N4002 protect the integrated circuit.

Figure 5.3 below shows some other examples of diodes. The life of a globe can be increased by adding a diode as shown in 5.3a. By simply connecting it in series, the current passing through the globe is halved and it lasts a lot longer. However the brightness is reduced and the light becomes yellow. The Diode should have a reverse voltage of over 400V, and a current higher than the globe. A 1N4004 or BY244 is suitable.

A very simple DC voltage stabilizer for low currents can be made using 5.3c as a reference.

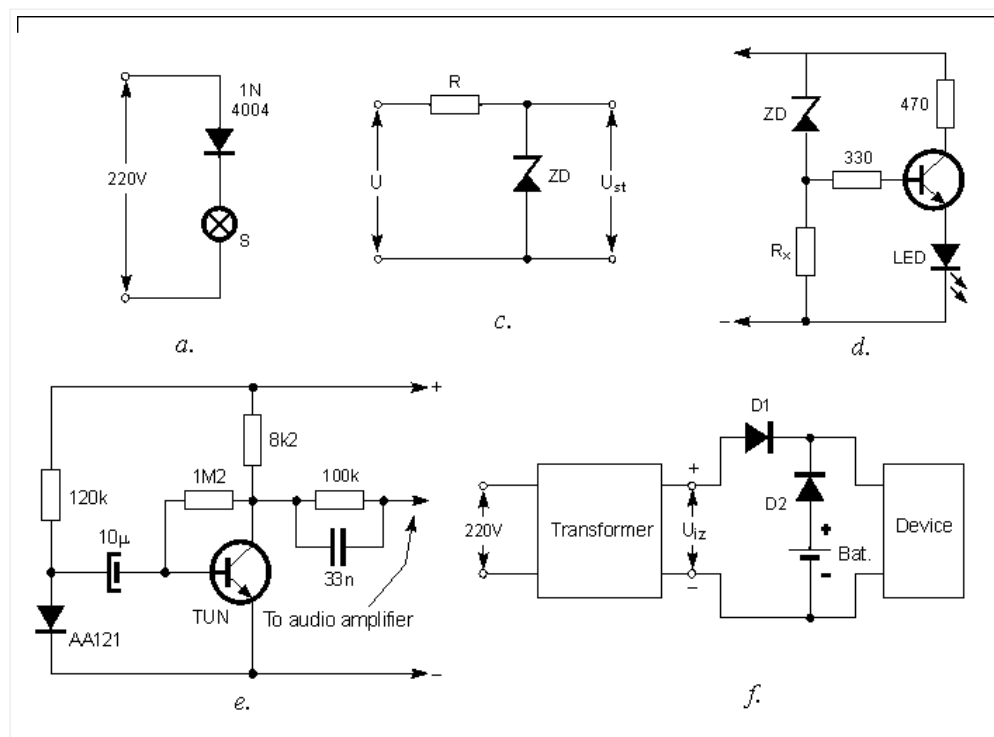


Fig. 5.3: a – using a diode to prolong the light bulb's life span, b – stair-light LED indicator, c – voltage stabilizer, d – voltage rise indicator, e – rain noise synthesizer, f – backup supply

Unstabilized voltage is marked "U", and stabilized with "UST." Voltage on the Zener diode is equal to U_{st} , so if we want to achieve a stabilized 9V, we would use a ZPD9.1 diode. Although this stabilizer has limited use it is the

basis of all designs found in power supplies.

We can also devise a voltage overload detector as shown in figure 5.3d. A LED indicates when a voltage is over a predefined value. When the voltage is lower than the operating voltage of the Zener, the zener acts as a high value resistor, so DC voltage on the base of the transistor is very low, and the transistor does not “turn on.” When the voltage rises to equal the Zener voltage, its resistance is lowered, and transistor receives current on its base and it turns on to illuminate the LED. This example uses a 6V Zener diode, which means that the LED is illuminated when the voltage reaches that value. For other voltage values, different Zener diodes should be used. Brightness and the exact moment of illuminating the LED can be set with the value of R_x .

To modify this circuit so that it signals when a voltage drops below some predefined level, the Zener diode and R_x are swapped. For example, by using a 12V Zener diode, we can make a car battery level indicator. So, when the voltage drops below 12V, the battery is ready for recharge.

Figure 5.3e shows a noise-producing circuit, which produces a rain-like sound. DC current flowing through diode AA121 isn't absolutely constant and this creates the noise which is amplified by the transistor (any NPN transistor) and passed to a filter (resistor-capacitor circuit with values 33nF and 100k).

Figure 5.3f shows a battery back-up circuit. When the “supply” fails, the battery takes over.



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